

AN EXACT RECURRENCE TRANSITION IN COLORED CATALAN RENEWAL SHIFTS

ANONYMOUS

ABSTRACT. For each integer $c \geq 1$ we build an aperiodic countable-state loop shift with exactly cC_{n-1} first-return loops of length n , where C_j is a Catalan number. Its first-return series is

$$F_c(z) = \frac{c}{2}(1 - \sqrt{1 - 4z}).$$

The single parameter c produces an exact three-way transition. The shift is transient at $c = 1$, null recurrent at $c = 2$, and strongly positive recurrent for $c \geq 3$. Its Gurevich entropy is

$$h_G(c) = \begin{cases} \log 4, & c = 1, 2, \\ \log(c^2/(c-1)), & c \geq 3, \end{cases}$$

and a measure of maximal entropy exists exactly when $c \geq 3$. In that regime its normalized first-return law and mean return time are explicit. The formal periodic-orbit zeta function is algebraic:

$$\zeta_c(z) = \frac{1}{1 - F_c(z)}.$$

At the two boundary parameters this yields the exact fixed-point laws $\#\text{Fix}(\sigma^n) = \frac{1}{2}\binom{2n}{n}$ for $c = 1$ and $\#\text{Fix}(\sigma^n) = 4^n/2$ for $c = 2$. Thus the recurrence transition is also visible as a change in the dominant zeta singularity.

1. INTRODUCTION

Countable Markov shifts can be transient, null recurrent, or positive recurrent even when their Gurevich entropy is finite. The classification by first-return series, and its consequences for maximal-entropy measures, are standard parts of the thermodynamic formalism of Gurevich, Savchenko, and Sarig [1, 3]. Loop and circular-code presentations also turn first-return generating functions into dynamical zeta functions [2]. These general mechanisms are prior input, not claims of this note.

Our purpose is to isolate a one-parameter laboratory in which every part of the classification is elementary and closed form. Catalan growth is exactly critical for this task: the first-return series has radius $1/4$, a finite boundary value, and an infinite boundary derivative. Multiplying its coefficients by c therefore separates transience, null recurrence, and positive recurrence at the consecutive integer values $1, 2, 3$.

The contribution is the complete computation for the colored Catalan loop family: entropy, recurrence class, existence and return law of the maximal measure, zeta singularity, and two exact fixed-point sequences. We do not claim to introduce renewal shifts, the Catalan generating function, the Vere–Jones classification, or circular-code zeta identities.

2. THE LOOP FAMILY

Write

$$C_j = \frac{1}{j+1} \binom{2j}{j}, \quad j \geq 0.$$

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Fix $c \in \mathbb{N}$. Construct a directed multigraph $\mathcal{G}_c$ with a distinguished vertex o as follows. For every $n \geq 1$, take

$$f_n(c) = cC_{n-1}$$

simple directed loops of length n based at o . Loops have disjoint internal vertices; the c loops of length one are parallel edges at o . Let (Σ_c, σ) be the two-sided edge shift of $\mathcal{G}_c$.

The graph is strongly connected, and its length-one loops make its period one. There are finitely many first-return loops of every fixed length, so all formal series below are coefficientwise well defined. The first-return series at o is

$$(1) \quad F_c(z) = \sum_{n \geq 1} f_n(c)z^n = cz \sum_{j \geq 0} C_j z^j = \frac{c}{2}(1 - \sqrt{1 - 4z}).$$

Its radius is $R = 1/4$, and

$$(2) \quad F_c(R) = \frac{c}{2}, \quad F'_c(z) = \frac{c}{\sqrt{1 - 4z}}.$$

Remark 2.1 (object boundary). This is a countable-state Markov shift, not a compact finite-alphabet subshift. Entropy below means Gurevich entropy. Statements about maximal entropy concern invariant Borel probability measures on Σ_c . For an edge sequence x , we write $[o]$ for the vertex cylinder on which the initial vertex of the edge x_0 is o .

3. ENTROPY AND THE RECURRENCE TRANSITION

For an irreducible loop shift, let $r_c = e^{-h_G(c)}$. The standard renewal criterion says that r_c is the smallest positive solution of $F_c(r) = 1$ when such a solution occurs no later than R ; otherwise $r_c = R$. At this parameter the system is transient if $F_c(r_c) < 1$, recurrent if equality holds, and a recurrent system is positive recurrent precisely when

$$(3) \quad \sum_{n \geq 1} n f_n(c) r_c^n = r_c F'_c(r_c) < \infty.$$

Theorem 3.1 (exact trichotomy). *For the colored Catalan loop shift:*

- (1) $c = 1$ is transient and $h_G(1) = \log 4$;
- (2) $c = 2$ is null recurrent and $h_G(2) = \log 4$;
- (3) every $c \geq 3$ is strongly positive recurrent, with

$$r_c = \frac{c-1}{c^2}, \quad h_G(c) = \log \frac{c^2}{c-1}.$$

An invariant probability measure of maximal entropy exists if and only if $c \geq 3$, and is then unique.

Proof. At $c = 1$, (2) gives $F_1(R) = 1/2 < 1$, hence transience and $r_1 = R$. At $c = 2$, one has $F_2(R) = 1$, but $F'_2(R) = \infty$; (3) gives null recurrence.

If $c \geq 3$, solving $F_c(r) = 1$ gives

$$\sqrt{1 - 4r} = 1 - \frac{2}{c}, \quad r = \frac{c-1}{c^2} < \frac{1}{4}.$$

The solution lies strictly inside the convergence disk, so $r F'_c(r) < \infty$ and the loop shift is positive recurrent. The strict gap $r < R$ is the loop-shift form of strong positive recurrence. Taking $-\log r$ gives the entropy formula. Existence and uniqueness of a maximal-entropy probability measure for an irreducible finite-entropy countable Markov shift are equivalent here to positive recurrence [1, 3]. Concretely, at $c = 1$ the candidate loop weights have total mass $F_1(R) < 1$; at $c = 2$ they have total mass one but infinite mean return time; and for $c \geq 3$ they have both total mass one and finite mean. These are exactly the normalization conditions for the stationary renewal probability. $\square$

The maximal measure can be written without eigenvectors. On each return to o , choose a particular first-return loop γ of length $|\gamma| = n$ with probability r_c^n . The normalization is exactly $F_c(r_c) = 1$.

Proposition 3.2 (explicit maximal measure). *Let $c \geq 3$. Under the unique maximal-entropy measure, the return length τ to o has law*

$$\mathbb{P}(\tau = n) = cC_{n-1}r_c^n, \quad n \geq 1,$$

and

$$\mathbb{E}\tau = r_c F'_c(r_c) = \frac{c-1}{c-2}, \quad \mu_c([o]) = \frac{c-2}{c-1}.$$

The return law has an exponential factor

$$4r_c = \frac{4(c-1)}{c^2} < 1.$$

Proof. There are cC_{n-1} loops of length n , each with probability r_c^n . The weights of all individual loops sum to $F_c(r_c) = 1$, and their mean length is finite by positive recurrence, so the associated stationary renewal tower is a probability measure. Its entropy per excursion is $(-\log r_c)\mathbb{E}\tau$; division by the mean excursion length gives entropy $-\log r_c = h_G(c)$, identifying it with the unique maximal measure. Differentiating (1) and using $\sqrt{1-4r_c} = (c-2)/c$ gives

$$r_c F'_c(r_c) = \frac{c-1}{c^2} \frac{c^2}{c-2} = \frac{c-1}{c-2}.$$

The stationary tower normalization, equivalently Kac's return identity, gives the reciprocal formula for $\mu_c([o])$. Finally $C_{n-1} = \Theta(4^{n-1}n^{-3/2})$, so the displayed strict inequality is the exponential tail factor. $\square$

4. PERIODIC ORBITS AND THE ALGEBRAIC ZETA FUNCTION

Every periodic path is a circular concatenation of first-return loops. The loop code is circular because distinct loops have private internal vertices. The usual loop-counting identity [2] therefore gives, as a formal power series,

$$(4) \quad \zeta_c(z) := \exp\left(\sum_{n \geq 1} \frac{\#\text{Fix}(\sigma^n)}{n} z^n\right) = \frac{1}{1 - F_c(z)} = \frac{1}{1 - \frac{c}{2} + \frac{c}{2}\sqrt{1-4z}}.$$

There are finitely many n -periodic points for every n : a periodic path of length n only uses first-return loops of length at most n .

Theorem 4.1 (boundary fixed-point laws). *At the transient and null-recurrent parameters,*

$$\zeta_1(z) = \frac{1 - \sqrt{1-4z}}{2z}, \quad \zeta_2(z) = (1-4z)^{-1/2},$$

and, for every $n \geq 1$,

$$\#\text{Fix}_{\Sigma_1}(\sigma^n) = \frac{1}{2} \binom{2n}{n}, \quad \#\text{Fix}_{\Sigma_2}(\sigma^n) = \frac{4^n}{2}.$$

For $c \geq 3$, ζ_c has a simple pole at $r_c = (c-1)/c^2$, strictly before its Catalan branch point $1/4$.

Proof. For $c = 1$, rationalizing (4) gives the Catalan generating function. The classical identity

$$[z^n] \log\left(\frac{1 - \sqrt{1-4z}}{2z}\right) = \frac{1}{2n} \binom{2n}{n}$$

and the definition of dynamical zeta give the first fixed-point law. For $c = 2$, taking the logarithm of $(1-4z)^{-1/2}$ gives $[z^n] \log \zeta_2 = 4^n/(2n)$. If $c \geq 3$, the denominator vanishes at $r_c < 1/4$, and its derivative there is $-F'_c(r_c) \neq 0$, proving the simple pole statement. Moreover, $|F_c(z)| \leq F_c(|z|) < 1$ for $|z| < r_c$. On $|z| = r_c$, equality $F_c(z) = 1$ in the triangle inequality forces the positive z and z^2 terms to have the same argument, hence $z = r_c$. Thus this pole is the unique singularity of smallest modulus. $\square$

The singularity change is sharper than entropy alone. The cases $c = 1$ and $c = 2$ have the same entropy $\log 4$, but at $c = 1$ the zeta function remains finite at $1/4$ and has a branch point, whereas at $c = 2$ it diverges with exponent $1/2$. At $c \geq 3$ a simple pole moves strictly inside the Catalan disk, exactly when the maximal-entropy probability measure appears.

5. CHECKS, OWNERSHIP, AND SCOPE

The accompanying deterministic control independently generates Catalan numbers, composes return loops, recovers fixed-point counts from the formal zeta identity, checks both boundary laws through order 40, and verifies the closed entropy and mean-return formulas for $3 \leq c \leq 50$. These finite checks support the algebra and guard indexing; the proofs above carry the all-parameter statements.

The general recurrence classification and maximal-measure criterion are owned by countable-state Markov-shift theory [1, 3]. The zeta reduction to a circular first-return code is also prior methodology [2]. The Catalan formula and its square-root singularity are classical. The narrow result here is their explicit synthesis for the family $f_n = cC_{n-1}$. No absolute novelty, priority, compactness, or finite-state claim is made.

REFERENCES

- [1] B. M. Gurevich and S. V. Savchenko. Thermodynamic formalism for countable symbolic markov chains. *Russian Mathematical Surveys*, 53(2):245–344, 1998. doi: 10.1070/RM1998v053n02ABEH000017.
- [2] Gerhard Keller. Circular codes, loop counting, and zeta-functions. *Journal of Combinatorial Theory, Series A*, 56(1):75–83, 1991. doi: 10.1016/0097-3165(91)90023-A.
- [3] Omri M. Sarig. Thermodynamic formalism for countable markov shifts. *Ergodic Theory and Dynamical Systems*, 19(6):1565–1593, 1999. doi: 10.1017/S0143385799146820.